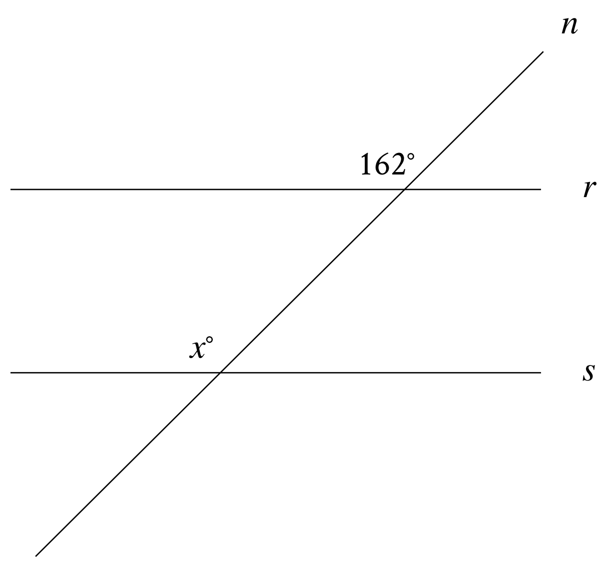
Question ID: 5b918ebb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure, line n intersects lines r and s. Line r is parallel to line s. What is the value of x?

Correct Answer: 162

Rationale

The correct answer is 162. It's given that line r is parallel to line s. Since line n intersects both lines r and s, it's a transversal. The angles in the figure marked as $162°$ and $x°$ are on the same side of the transversal, where one is an interior angle with line s as a side, and the other is an exterior angle with line r as a side. Thus, the marked angles are corresponding angles. When two parallel lines are intersected by a transversal, corresponding angles are congruent and, therefore, have equal measure. It follows that the value of x is 162.